বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা

BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

EXAMINATION CONFIDENTIAL

Term Final Examination, SPRING 2025

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE 223

Course Title: Digital Logic Design

Credit Hour: 03

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Each question carries **30** marks.
b. **Answer all of the following FIVE questions.**
c. The figure on the right of each question indicates marks for the respective question.
d. Use a different answer script for each part.

The Course Outcomes (COs) are:

CO1: Remember and understand the number system and Boolean algebra and basic properties Boolean algebra to simplify simple Boolean functions.

CO2: Understanding and applying the tabulation and Karnaugh map methods for simplify combinational circuits.

CO3: Identify the basic sequential logic components: SR Latch, Different Flip-Flops and their use and able to analyze sequential logic circuits.

Part-A

1. **a.** Design a two-bit parrel adder circuit and show the calculation if the value of A=10, B=11. **[CO2]** **20**

   b. Design a combinational circuit with three inputs and one output. The output is 1 when the binary value of the inputs is an odd number. **[CO2]** **10**

   OR

   Design a combinational circuit with three inputs and one output. The output is 1 when the binary value of the inputs is less than 3. The output is 0 otherwise. **[CO2]**

2. **a.** What is decoder? Construct a **5-to-32-line decoder** with **four 3-to-8-line decoders** with enable and **a 2-to-4-line decoder**. Use block diagrams for the components. **[CO2]** **15**

   OR

   Design an **excess-3-to-binary decoder** using the unused combinations of the code as don't care conditions. **[CO2]**

b. Write a short note on the following terms: **[CO2]** 15

i) PROM

ii) EPROM

iii) PLA

3. a. Define combinational logic. How does it differ from sequential logic? **[CO2]** 10

b. Design a **5-1** Multiplexer and **1-4** Demultiplexer circuit. **[CO2]** 10

c. Implement the following Boolean functions using PLA(Programmable Logic Array): 10

$$F1 = AB+A'C$$

$$F2 = A'B+BC$$

The PLA has 3 inputs, 4 product terms, and 2 outputs. Draw the PLA programming table and explain how the logic functions are realized. **[CO2]**

Part-B

4. a. With proper diagram explain the working principle of Full Adder & Full Subtractor. **[CO3]** 10

b. Explain the operation of a **Clocked SR** flip-flop and a **D-type** flip-flop, using their respective logic diagrams. **[CO3]** 10

c. Draw the **T type** flip-flop's logic diagram and create its characteristic table and equation. **[CO3]** 10

5. a. Explain the role of Registers and Counters. **[CO3]** 10

b. Design and analyze a clocked sequential circuit. Based on the circuit, derive the state table, construct the corresponding state diagram, and formulate the state equations. **[CO3]** 20

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 2025

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: GED-221

Course Title: Engineering Economics

Credit Hour: 03

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Figure on the right of each question indicates marks for the respective question.

b. Answer ***any FIVE*** questions out of **SIX**.

c. Use different answer scripts for each section

d. Course Learning Outcomes are:

CLO1: Develop preliminary knowledge of economic concepts for understanding of national and international economy

CLO2: Understand how markets work, when markets fail and how the government can affect market outcomes

CLO3: Evaluate macroeconomic policies based on their effects on the economic variables of interest

CLO4: Apply different techniques of investment decisions.

PART-A

A1. i. Define the concept of utility? Justify "Ordinal utility is more realistic than cardinal utility". **[CLO1]** [15]

ii. Explain the law of diminishing marginal utility with a hypothetical example. **[CLO1]** [15]

A2. i. Explain the concept of Breakeven Point with diagram. **[CLO1]** [4]

ii. Show the derivation of savings function from consumption function with the help of diagram. **[CLO1]** [10]

iii. A savings function is given as, $S = -a + (1-b)Y$ **[CLO2]** [16]

a) Find out the consumption function.
b) Show that, APS+APC= 1
c) Show that, MPS+MPC=1
d) Find out the breakeven point.

PART-B

B3. i. Explain different types of returns to scale. **[CLO1]** [15]

ii. A production function is given as: $Q = 10L^{0.4}K^{0.6}$. Where, Q = Output, L=Labor and K= Capital. The firm employs 16 units of labor and 25 units of capital. **[CLO4]** [15]

a) Calculate the marginal product of labor (MPL) at this point.
b) Interpret the economic meaning of the marginal product of labor in this context.
c) If the firm increases labor to 25 units (keeping capital fixed at 25), does the marginal product of labor increase or decrease? Justify using appropriate calculations or reasoning.

If the firm doubles both labor and capital (L=32, K=50), what happens to output? Does this production function exhibit increasing, constant, or decreasing returns to scale? Justify your answer with calculations.

B4. i. Explain the factors of production. **[CLO1]** [15]

ii. Differentiate capital intensive production function from labor intensive production technique. **[CLO1]** [15]

B5. i. Explain the concept of demand-pull inflation with appropriate reasons and diagram. **[CLO1]** [10]

ii. Suppose a consumer's average spending on a basket of goods includes three items: rice, chicken, and wheat. The prices of rice, chicken, and wheat in 2020 (base year) are TK 70, TK 180 and TK 40 per kg respectively. The prices of rice, chicken, and wheat in 2024 (current year) are TK 80, TK 190 and TK 50 per kg respectively. Calculate the Consumer price index (CPI). Interpret the result. **[CLO2]** [10]

iii Describe the fiscal and monetary policy instruments to reduce inflation rate in an economy. **[CLO3]** [10]

B6. i. Explain the concept of Indifference curve. Write down the properties of Indifference curve. **[CLO1]** [15]

ii. Suppose, a consumer has four combinations of consuming two goods (Q_1, Q_2). He/she has the right to choose any combination. Find out the $MRS_{A\ to\ B}, MRS_{B\ to\ C}, MRS_{C\ to\ D}$ from the following table and interpret the result. **[CLO1]** [15]

Combination	Quantity of Good 1 (Q_1)	Quantity of Good 2 (Q_2)
A	80	10
B	60	20
C	30	40
D	20	70

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 2025

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE-225

Course Title: Numerical Methods

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. Answer ***any FIVE*** questions out of ***FIVE***.
c. Use different answer scripts for each section.
d. Course Learning Outcomes are

CLO1: Recall the basics of accuracy and finite precision, round-off errors, truncation errors, and stability of numerical algorithms.
CLO2: Apply appropriate numerical techniques and calculate the computational cost in the fields of science and engineering.
CLO3: Formulate numerical techniques for root finding, curve fitting, differentiation, and integration.

Part-A

1. a. The cost function of producing items is $C(x) = x^2 + 10x + 100$. Apply forward finite difference to estimate the marginal cost when $x = 5$, using $h = 1$. **[CO3→C3]** 15

b. The population (in thousands) of a city is given at different years: 15

Year	2000	2005	2010	2015	2020
population	65.2	70.3	76.5	83.1	95.5

Use Newton's backward difference interpolation to determine the population in the year 2012. **[CO3→C3]**

OR

Using an appropriate formula for interpolation, estimate the number of students who obtained marks between 40 and 45 from the following table: **[CO3→C3]**

Marks	30-40	40-50	50-60	60-70	70-80
No. of Students	31	42	51	35	31

Why transformation is needed in Gauss Quadrature formula? Use the two-point Gauss quadrature to approximate the value of the integral: $\int_0^1 x^2 e^{-x}\, dx$. **[CO3→C3]**

2. a. Solve the following linear system with Gauss Elimination method. **[CO3→C3]** 15

$$x + y + z = 9$$
$$2x - 3y + 4z = 13$$
$$3x + 4y + 5z = 40$$

b. Given the system: 15

$$4x - y + z = 7$$
$$x + 3y - z = 4$$
$$x - y + 5z = 6$$

Analyze whether Jacobi method will converge or not. **[CO3→C4]**

3 a. Applying Simpson's 1/3 rule to estimate the integral: $\int_0^6 \frac{1}{1+x^2}\,dx$, take 6 intervals. 15
[CO3→C3]

b. Apply the Trapezoidal rule to approximate: $\int_0^4 x^2\,dx$, using 4 intervals. **[CO3→C3]** 15

Part-B

4. a. Solve the following linear system with Gauss Seidel Iteration method. **[CO3→C3]** 15

$$27x + 6y - z = 85$$
$$6x + 15y + 2z = 72$$
$$x + y + 54z = 110$$

b. Apply Gauss Jordan method, solve the following equations. **[CO3→C3]** s 15

$$3x - y + z = 4$$
$$x + y + z = 2$$
$$2x + 3y + 4z = 8$$

5 a. Find the Taylor series of $x^3 - 10x + 6$ at $x = 3$. **[CO3→C3]** 10

b. Consider an ordinary differential equation $\frac{dy}{dx} = x^2 + y^2$; $y(1) = 1.2$ Find the value of $y(1.05)$ at $h = 0.05$. **[CO3→C3]** 20

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 2025
Department of Computer Science and Engineering
Level-2 Term-II
Course Code: CSE-221
Course Title: Computer Algorithms and Complexity
Credit Hour: 03
Exam Duration: 3 Hours
Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. Answer ***all FIVE*** questions.
c. Use different answer scripts for each part.
d. Course Learning outcomes are:

CLO1: Analyze the asymptotic performance and write rigorous correctness proofs for algorithms.
CLO2: Estimate the performance of various algorithms and data structures.
CLO3: Apply important algorithmic design paradigms and methods of analysis for problem solving.
CLO4: Device efficient algorithms in common engineering design situations.

Part-A

1. a. "Bellman-Ford algorithm is preferred over Dijkstra's algorithm in graphs containing negative edge weights". Explain and Justify your explanation with an example. **[CO2→C2]** **10**

OR

Write the pseudocode for the Bellman-Ford algorithm to find the shortest paths from a single source vertex to all other vertices in a weighted directed graph (which may contain negative weight edges but no negative weight cycles). **[CO2→C4]**

b. Explain the concept of optimal substructure. Provide one example problem that can be solved using both Dynamic Programming and Greedy approaches. Justify which approach gives the optimal result and why. **[CO2→C2].** **10**

c. Write the pseudocode for the Longest Common Subsequence (LCS) algorithm using dynamic programming. **[CO2→C3]** **10**

2. a. You are given a bag with a weight capacity of 8 units. You have the following items, each with a weight and a value: **10**

Item	Weight (units)	Value (units)
A	2	1
B	3	2
C	4	5
D	5	6

Apply the 0/1 knapsack algorithm to find the maximum value and items you can carry in the knapsack. **[CO3→C3]**

OR

You are given a knapsack with a weight capacity of 15 units. You have the following items, each with a weight and a value:

Item	Weight (units)	Value (units)
A	2	10
B	3	5
C	5	15
D	7	7
E	1	6
F	4	18
G	1	3

Apply the fractional knapsack algorithm to find the maximum value you can carry in the knapsack? **[CO3→C3]**

b. You are given the following set of jobs, each with a deadline and a profit. Find the maximum profit using Job Sequencing with Deadline algorithm. **[CO3→C3]** **10**

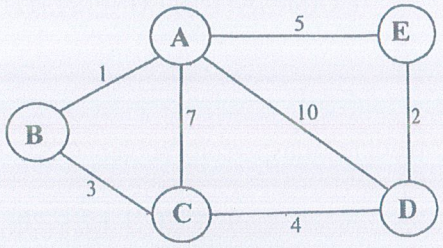
Job	Profit	Deadline
J_1	35	3
J_2	30	4
J_3	25	4
J_4	20	2
J_5	15	3
J_6	12	1
J_7	5	2

c. Consider the following message "AAABBCDDDEEFFGGHHIIJJ". Draw the tree and assign binary codes to each character using Huffman Coding. **[CO3→C3]** **10**

OR

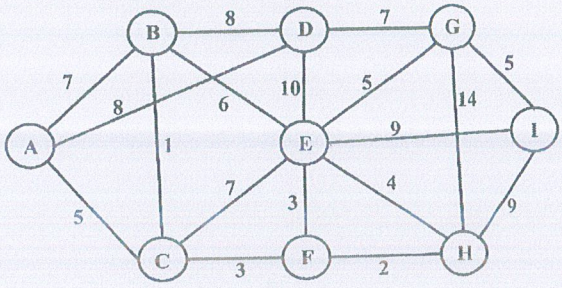
Given two sequences, X= "COMPUTER" and Y= "COMMUNITY". Find the length and sequence of the longest common subsequence for X and Y. **[CO3→C3]**

3. a. 10



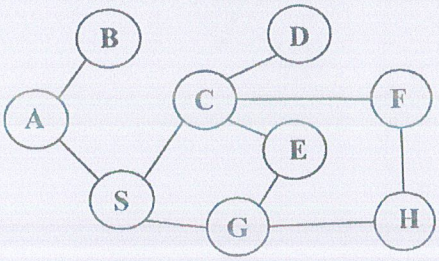
Illustrate the step-by-step execution of Dijkstra's algorithm on a given weighted graph, indicating how the MST is constructed. **[CO3→C3]**

b. Given the following connected, undirected, weighted graph with a set of vertices and edges, apply both **Kruskal's and Prim's algorithms** to generate the Minimum Spanning Tree (MST). For each algorithm, clearly show the intermediate steps involved in the edge/vertex selection process. **[CO3→C3]** 20

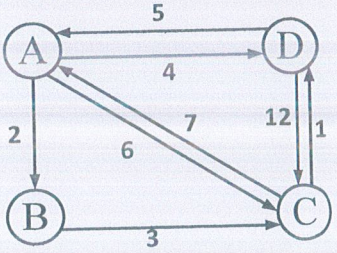


Part-B

4. a. Perform both Breadth First Search (BFS) and Depth First Search (DFS) algorithms starting from vertex A. List the order in which the vertices are visited during the traversal. **[CO3→C3]** 15



b. You are given the following weighted directed graph. Apply the Floyd–Warshall algorithm to compute the shortest paths between all pairs of vertices. **[CO3→C3]** 15



5. a. Given the following adjacency matrix for 4 cities, manually compute the minimum cost to complete the Travelling Salesperson Problem (TSP) tour using the dynamic approach: **[CO4→C4]** **12**

0	10	15	20
10	0	35	25
15	35	0	30
20	25	30	0

b. Using the backtracking algorithm, find all possible solutions for placing 8 queens on an 8×8 chessboard so that no two queens threaten each other. **[CO4→C4]** **18**

17/08/25
G-B

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 2025

Department of Computer Science and Engineering

Level-2 Term-II
Course Code: MATH -221
Course Title: MATH IV (Probability and Statistics)
Credit Hour: 03
Exam Duration: 03 hours
Full Marks -150

Notes:

a. Each question carries 30 marks.
b. Answer any five (05) questions. **Either three from Part-A and two from Part-B, or two from Part-A and three from Part-B.**

Part-A

1. a. A fire brigade has two fire engines operating independently. The probability that a specific fire engine is available when needed is 0.99. **12**

i) **Determine** the probability that an engine is available when needed.

ii) **Find** the probability that neither is available when needed.

b. Suppose a sample space consists of 500 persons and are distributed according to their sex and employment status as shown in the accompanying table. **18**

	Employment Status		
Sex	Employed (E)	Unemployed (U)	Total
Male (M)	255	20	275
Female (F)	80	145	225
Total	335	165	500

Compute the following probabilities:

i) the selected person is a male
ii) the selected person is female and unemployed
iii) the selected person is male and unemployed
iv) female and not unemployed.

2. a. **Find** whether the random variable X and Y are independent if their joint density function is : **12**

$$f(x,y) = 4xy, \quad \text{for } 0 < x < 1,\ 0 < y < 1$$
$$= 0, \quad \text{otherwise}$$

b. Suppose X and Y have the following joint probability distribution **18**

	X				
Y	0	1	2	3	Row sum
0	0	1/8	2/8	1/8	4/8
1	1/8	2/8	1/8	0	4/8
Column sum	1/8	3/8	3/8	1/8	1

Estimate the marginal distribution of X and Y.

3. a. Let X designate the number of heads obtained in the toss of three ideal coins. If two or more heads turn up, you receive Tk.10; otherwise you pay Tk.5. **15**
Determine

i) the expected number of heads to turn up.
ii) the expected amount of money you would win.

b. **Calculate** the marginal distributions of X and Y from the following joint probability function. **15**

$$f(x,y) = \frac{1}{8}(6 - x - y), \quad \textit{for } 0 < x < 2,\ 2 < y < 4$$
$$= 0, \quad \textit{elsewhere}$$

Also **obtain** $P(X + Y < 3)$ and $P(X < 1.5, Y < 2.5)$

Part-B

1. a. **Express** the two regression equations of X on Y and Y on X from the data given below, taking deviations from an actual means of X and Y : **18**

Price (Rs.)	10	12	13	12	16	15
Amount demanded	40	38	43	45	37	43

b. **Determine** the means of X and Y variables and the coefficient of correlation between them from the following two regression equations : **12**

$$2Y - X - 50 = 0$$
$$3Y - 2X - 10 = 0.$$

2. a. **Estimate** the Karl pearson's coefficient of Skewness and measure of Kurtosis based on moment of the given data and **describe** the result. **18**

Investment ($)	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	12	18	20	15	10	3	2

b. **Find** the Skewness using Keyley's coefficient and Bowley's system for the given distribution **12**

Class Interval	50 – 55	55 – 60	55 – 60	65 – 70	70 – 75	75 – 80	80 – 85
Frequency	17	15	26	32	22	14	9

3. a. A statistics teacher wants to compare his two classes to see if they performed any differently on the tests he gave that semester. Class A had 25 students with an average score of 70, $S.D = 15$. Class B had 20 students with an average score of 74, $S.D = 25$. Using level of significance 0.01 and 0.05, **prove** that these two classes perform differently or same on the tests. **15**

b. A company wants to know if the preference for three different product designs (A, B, C) is the same across three different regions (North, East, West). They surveyed 100 people from each region. **Conduct** the test of hypothesis for homogeneity at a 5% significance level $(\alpha = 0.05)$. **15**

Product Design / Region	North	East	West	Total
Design A	40	30	35	105
Design B	35	45	30	110
Design C	25	25	35	85
Total	100	100	100	300

t-test table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646

Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of x^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38